UNIVERSIDADE FEDERAL DE SANTA CATARINA
CAMPUS REITOR JOÃO DAVID FERREIRA LIMA
POSTGRADUATE PROGRAM IN AUTOMATION AND SYSTEMS ENGINEERING

Vitor Della Torre dos Santos

BVOLT:
A FIVE-CELL BATTERY BANK MICROGRID DIGITAL TWIN COMPONENT

Florianópolis, Santa Catarina – Brazil
2019

Vitor Della Torre dos Santos

BVOLT:
A FIVE-CELL BATTERY BANK MICROGRID DIGITAL TWIN COMPONENT

Master's Dissertation submitted to the Postgraduate Program in Automation and Systems Engineering of Universidade Federal de Santa Catarina for degree acquirement in Master in Automation and Systems Engineering.
Supervisor: Julio Elias Normey-Rico, Phd.
Co-supervisor: Diogo Ortiz Machado, Phd.

Florianópolis, Santa Catarina – Brazil
2019

Legal Notes:

There is no warranty for any part of the documented software. The authors have taken care in the preparation of this thesis, but make no expressed or implied warranty of any kind and assume no responsibility for errors or omissions. No liability is assumed for incidental or consequential damages in connection with or arising out of the use of the information or programs contained here.

Cataloging at source by the University Library of the Federal University of Santa Catarina.
File compiled at 09:32h of the day Friday 20th February, 2026.

Vitor Della Torre dos Santos
BVOLT: A Five-Cell Battery Bank Microgrid Digital Twin Component / Vitor Della Torre dos Santos; Supervisor, Julio Elias Normey-Rico, Phd.; Co-supervisor, Diogo Ortiz Machado, Phd. – Florianópolis, Santa Catarina – Brazil, 25 of februrary of 2026.
57 p.

Master's Dissertation – Universidade Federal de Santa Catarina, EAS – Department of Automation and Systems Engineering, CTC – Technological Center, Postgraduate Program in Automation and Systems Engineering.

Includes references

1. Keyword 1, 2. Keyword 2, 3. Keyword 3, I. Julio Elias Normey-Rico, Phd. II. Diogo Ortiz Machado, Phd. III. Postgraduate Program in Automation and Systems Engineering IV. BVOLT: A Five-Cell Battery Bank Microgrid Digital Twin Component

CDU 02:141:005.7

Vitor Della Torre dos Santos

BVOLT:
A FIVE-CELL BATTERY BANK MICROGRID DIGITAL TWIN COMPONENT

This Master's Dissertation was considered appropriate to get the Master in Automation and Systems Engineering, and it was approved by the Postgraduate Program in Automation and Systems Engineering of EAS – Department of Automation and Systems Engineering, CTC – Technological Center of Universidade Federal de Santa Catarina.

Florianópolis, Santa Catarina – Brazil, 25 of februrary of 2026.

Julio Elias Normey-Rico, Phd.
Coordinator of Postgraduate Program in
Automation and Systems Engineering

Examination Board:

Julio Elias Normey-Rico, Phd.
Supervisor
Universidade Federal de Santa
Catarina – UFSC

Diogo Ortiz Machado, Phd.
Co-supervisor
Universidade Federal de Santa
Catarina – UFSC

Leandro Buss Becker, Phd.
Universidade Federal de Santa Catarina –
UFSC

This work is dedicated to uncle Eduardo,
Who told be not everything has to be forged in iron and fire.

ACKNOWLEDGEMENTS

This work is a composition of my own will with the strength given to me by a group of amazing people—who lent me their power when I most needed. I want to thank my parents Alexandre and Márcia, who raised me through my whole path with unconditional love and support. I want to thank my brother Thales, who has shown me what hard work looks like when one strives for their passions. And I want to thank Luana, my beloved and bestfriend, who made me wake up in the morning feeling alive and grateful even when my life was about to crumble and blow away. I also want to thank professors Julio and Diogo for everything they have done to me during this whole process—I know things were not easy, but they would have been impossible without this team.

I would not be here entirely as I am now without the help of these people.

“I wouldn’t try that! I’m...”
INVINCIBLE

RESUMO

A integração de Recursos Energéticos Renováveis (RERs) é fundamental para a descarbonização, sendo as microrredes (MGs) uma solução essencial para gerenciar a intermitência e os problemas de qualidade de energia associados às fontes renováveis. Embora os Gêmeos Digitais de Microrredes (MGDTs) sejam capazes de lidar com as complexidades da integração de MGs, o desenvolvimento desses modelos de simulação e controle é frequentemente demorado, suscetível a erros e resulta em artefatos personalizados e pouco generalizáveis. Este trabalho propõe BVOLT: um software modular e generalizável projetado para simular, modelar e controlar componentes de microrredes. O objetivo principal do projeto é criar uma interface flexível e de fácil utilização entre MGDTs e usuários finais, a fim de aprimorar testes e interações. Como caso de testes, propõe-se um Gêmeo Digital de um banco de baterias de lítio composto por 5 células como componente de MGDT, visando validar a criação e a verificação de Gêmeos Digitais de Microrredes por meio de modelos baseados em princípios físicos e orientados a dados. Espera-se que o BVOLT se consolide como uma plataforma de referência no desenvolvimento de componentes de MGDT, atuando como um elo eficaz entre pesquisadores e equipamentos físicos.

Palavras-chaves: Microrredes, Recursos Energéticos Renováveis, Gêmeos Digitais

RESUMO EXPANDIDO

INTRODUÇÃO

O aumento das emissões de gases de efeito estufa, especialmente metano (CH_4), óxido nitroso (N_2O) e dióxido de carbono (CO_2), tem provocado elevação significativa e sem precedentes da temperatura média global. Esse cenário resulta em secas, enchentes, elevação do nível do mar, derretimento de calotas polares, perda de biodiversidade e ondas de calor prolongadas, com impactos potencialmente irreversíveis para os ecossistemas e para a sociedade. Diante desse contexto, as energias renováveis surgem como alternativa fundamental para a descarbonização, impulsionando esforços internacionais voltados à modernização dos sistemas de distribuição e gestão de energia. Entretanto, os recursos energéticos renováveis são inerentemente intermitentes e, combinados a perfis de carga variáveis, podem gerar desequilíbrios energéticos e problemas de qualidade de energia. Nesse cenário, as microrredes (MGs) se destacam como solução promissora, pois integram geração distribuída, sistemas de armazenamento e conversores de potência em redes de baixa tensão capazes de operar conectadas à rede principal (on-grid) ou de forma isolada (off-grid). Contudo, a integração de microrredes a sistemas centralizados é complexa e demanda modelos de simulação avançados, como os Gêmeos Digitais (DTs), capazes de representar com precisão os componentes físicos no ambiente virtual.

Os Gêmeos Digitais, originalmente desenvolvidos na indústria aeroespacial, dependem de fluxo bidirecional de dados e podem se basear em modelos físicos, orientados a dados ou híbridos. No contexto de microrredes, os Gêmeos Digitais de Microrredes (MGDTs) contribuem para aumentar a eficiência energética, estimar vida útil, reduzir custos e aprimorar monitoramento, previsão, detecção de falhas, controle, tomada de decisão e segurança cibernética. No entanto, o desenvolvimento desses modelos e de suas implementações computacionais é complexo, demorado e sujeito a erros.

Com o objetivo de acelerar a implantação de MGDTs e reduzir retrabalho, este trabalho propõe o desenvolvimento de um ambiente modular e generalizável. Como prova de conceito, apresenta-se a criação de um componente de Gêmeo Digital para um banco de cinco baterias de lítio UNICOBA UPLFP48, inserido na microrrede do LabInovação do INCT-CAPE. O INCT é uma iniciativa nacional voltada à pesquisa, formação de

recursos humanos e inovação em diversas áreas do conhecimento, incluindo energia e engenharia, sendo o INCT-CAPE dedicado ao controle e automação de processos energéticos.

O projeto BVOLT é apresentado como prova de conceito para o desenvolvimento de componentes de MGDT, com foco específico em armazenamento de energia. O trabalho descreve os materiais e métodos do laboratório, apresenta uma revisão de literatura sobre Gêmeos Digitais aplicados a microrredes e propõe o desenvolvimento do software para o banco de baterias, detalhando hipóteses, objetivos, cronograma e possíveis veículos de publicação científica.

OBJETIVOS

O objetivo geral deste trabalho é criar um Gêmeo Digital (DT) de um banco de baterias de lítio composto por 5 células, com o propósito de testar componentes de Gêmeos Digitais de Microrredes (MGDTs) e facilitar a criação de novos MGDTs por outros pesquisadores ao redor do mundo.

Os objetivos específicos concentram-se na validação de modelos baseados em princípios físicos e orientados a dados para o Gêmeo Digital do banco de baterias de 5 células.

- Tornar o banco de dados InfluxDB existente acessível remotamente, a fim de permitir curadoria de dados online e operação do Gêmeo Digital;
- Tornar o dashboard existente no Grafana habilitado para a nuvem, permitindo desenvolvimento remoto e uso mundial por outras equipes de pesquisa;
- Construir uma interface de software robusta, com funcionalidades plug-and-play, para a criação de um Gêmeo Digital de um banco de 5 baterias como componente de MGDT;
- Estimar o estado de carga (SOC) do banco de baterias de lítio de 5 células utilizando um modelo baseado em princípios físicos;
- Estimar o estado de carga (SOC) do banco de baterias de lítio de 5 células utilizando um modelo orientado a dados;

- Realizar uma comparação direta (head-to-head), definindo e executando experimentos comparativos entre os modelos físico e orientado a dados sob os mesmos cenários de teste;
- Integrar e implantar os estimadores escolhidos na pilha de Gêmeo Digital do B-VOLT, com ingestão de dados via Modbus para o InfluxDB, geração de dashboards no Grafana com curvas de SOC em tempo real, e garantir que a inferência possa ser executada no dispositivo-alvo (edge ou servidor); e
- Publicar o código, os conjuntos de dados e os resultados no GitHub, com notebooks reprodutíveis capazes de recriar as figuras principais, além da submissão de um artigo descrevendo o estudo comparativo realizado.

METODOLOGIA

Esta seção descreve os componentes físicos e digitais da microrrede do LabInovação e como suas conexões são estabelecidas. Os ativos físicos do laboratório são divididos em duas plantas: solar-hidrogênio e solar-térmica. Este projeto concentra-se exclusivamente na planta solar e de hidrogênio, uma vez que a planta solar térmica ainda se encontra em fase de projeto. Na planta solar-hidrogênio, o banco de baterias, os painéis fotovoltaicos, o emulador de gerador elétrico e o emulador eólico estão conectados ao inversor principal, que por sua vez está ligado à rede elétrica. Um segundo inversor, configurado em esquema pai-filho, conecta-se tanto ao primeiro inversor quanto a outro conjunto de painéis fotovoltaicos, sendo também responsável pelo gerenciamento dos equipamentos relacionados ao hidrogênio. A comunicação lógica entre o computador principal e o inversor principal ocorre via rede Modbus.

O computador principal executa um código-fonte desenvolvido integralmente em Python, estruturado segundo o paradigma de Programação Orientada a Objetos (POO) e organizado em ambiente virtual UV. Os dados são armazenados como séries temporais em um banco InfluxDB e visualizados por meio de dashboards no Grafana. O desenvolvimento do software segue os princípios da metodologia SOLID, visando garantir manutenibilidade, flexibilidade e escalabilidade. O sistema de software executa quatro serviços principais: (i) interface de comunicação via Modbus para leitura e escrita em registradores dos dispositivos; (ii) controle baseado em Model Predictive

Control (MPC) para regulação do estado de carga (SOC) do banco de baterias; (iii) previsão meteorológica para suporte às decisões operacionais; e (iv) gerenciamento de dados por meio de conexão com o InfluxDB.

A arquitetura do repositório segue um padrão limpo e instalável baseado em `src/` e `pyproject.toml`, com separação clara de responsabilidades. Há uma camada de abstração de dispositivos, adaptação de protocolo Modbus, armazenamento desacoplado em InfluxDB, agregação e normalização de telemetria, subsistema de otimização com modelos da microrrede e MPC, além de uma camada leve de apresentação. O projeto inclui ainda diretórios auxiliares para dados experimentais, documentação técnica e testes, formando um pipeline coerente que integra aquisição de campo, armazenamento, análise e controle.

RESULTADOS E DISCUSSÃO

Este trabalho parte do contexto macro da descarbonização energética, destacando que a crescente integração de recursos energéticos renováveis impõe desafios técnicos relacionados à intermitência, estabilidade e qualidade de energia. As microrredes emergem como solução estratégica para gerenciar geração distribuída, armazenamento e cargas dinâmicas, mas sua integração eficiente exige modelos de simulação avançados—especialmente Gêmeos Digitais, capazes de criar representações virtuais de alta fidelidade com fluxo bidirecional de dados. A revisão de literatura consolida o estado da arte em MGDTs, abordando:

- Modelagem física, orientada a dados e híbrida;
- Aplicações em renováveis, armazenamento, eletrônica de potência e infraestrutura de rede;
- Funções centrais como previsão, controle ótimo (incluindo MPC), resiliência e cibersegurança;
- Desafios estruturais, incluindo gestão de dados, interoperabilidade, padronização, segurança, custo computacional e escalabilidade.

Dentro desse panorama, o documento argumenta que ainda falta um ambiente modular e padronizado que facilite a construção, validação e comparação de componentes de MGDT. Para responder a essa lacuna, propõe-se o BVOLT, uma arquitetura de software que integra comunicação Modbus, armazenamento em InfluxDB, visualização via Grafana e um ecossistema Python estruturado segundo princípios de arquitetura limpa (SOLID).

Como prova de conceito, o trabalho desenvolve um Gêmeo Digital de um banco de 5 baterias de lítio, inserido na microrrede do LabInovação (INCT-CAPE). O núcleo técnico da proposta consiste na comparação direta entre dois estimadores de Estado de Carga (SOC):

- Um modelo baseado em princípios físicos (formulação em espaço de estados);
- Um modelo orientado a dados (aprendizado a partir de séries temporais multivariadas).

Ambos operam sobre a mesma infraestrutura de dados, permitindo comparação metodológica sob condições idênticas. A arquitetura do sistema conecta aquisição física, banco de dados, estimadores paralelos, benchmarking, ambiente de digital twin, API REST e dashboard remoto, formando um pipeline completo de aquisição, modelagem, controle e visualização.

Desta forma, a contribuição do trabalho não se limita à estimação de SOC, mas propõe um framework replicável para desenvolvimento de componentes de MGDT, com foco em modularidade, reprodutibilidade científica e acesso remoto. BVOLT é posicionado como um elo entre pesquisa acadêmica e infraestrutura física real, buscando acelerar a maturidade tecnológica de Gêmeos Digitais aplicados a microrredes.

CONSIDERAÇÕES FINAIS

Este trabalho apresentou o BVOLT como uma proposta modular para o desenvolvimento e validação de componentes de Gêmeos Digitais de Microrredes (MGDTs), tendo como prova de conceito um banco de cinco baterias de lítio inserido na microrrede do LabInovação (INCT-CAPE). A motivação central esteve associada aos desafios técni-

cos impostos pela integração de Recursos Energéticos Renováveis, cuja intermitência e variabilidade exigem ferramentas avançadas de modelagem, previsão e controle para garantir estabilidade, eficiência e resiliência energética. A revisão de literatura evidenciou que, embora os MGDTs representem uma das tecnologias mais promissoras no contexto da Indústria 4.0 aplicada a sistemas de energia, sua implementação ainda enfrenta barreiras significativas relacionadas à complexidade de modelagem, interoperabilidade, padronização, segurança cibernética, escalabilidade e custos computacionais. Nesse cenário, a ausência de ambientes generalizáveis e reutilizáveis limita a disseminação prática dessa tecnologia. Como resposta a essa lacuna, o BVOLT foi concebido como uma arquitetura limpa, extensível e orientada a serviços, integrando comunicação Modbus, armazenamento de séries temporais em InfluxDB, visualização via Grafana e modelos de estimação implementados em Python sob princípios SOLID. A abordagem dual—baseada em modelos físicos e modelos orientados a dados—permitiu não apenas a estimação do Estado de Carga (SOC) do banco de baterias, mas também a comparação metodológica direta entre paradigmas de modelagem, contribuindo para a discussão sobre interpretabilidade, adaptabilidade e robustez em aplicações de Gêmeos Digitais. Os resultados esperados incluem a consolidação de uma infraestrutura replicável para desenvolvimento de MGDTs, com acesso remoto, versionamento aberto e reprodutibilidade científica. A disponibilização de código, dados e experimentos comparativos reforça o compromisso com a transparência e com a construção colaborativa do conhecimento. Como perspectivas futuras, destacam-se a expansão do framework para outros ativos da microrrede (como sistemas fotovoltaicos, hidrogênio e eletrônica de potência), a exploração de modelos híbridos que combinem vantagens físicas e orientadas a dados, e a integração de estratégias avançadas de controle e cibersegurança. Dessa forma, o BVOLT pode evoluir de uma prova de conceito para uma plataforma de referência no desenvolvimento de componentes de Gêmeos Digitais aplicados a sistemas energéticos distribuídos.

Palavras-chaves: Microrredes, Recursos Energéticos Renováveis, Gêmeos Digitais

ABSTRACT

The integration of renewable energy resources (RERs) is critical for decarbonization, with microgrids (MGs) offering a key solution to manage the intermittency and power quality issues associated with renewables. While Microgrid Digital Twins (MGDTs) are capable of resolving the complexities of MG integration, the development of these simulation and control models is often time-costly, error-prone, and results in non-generalizable, customized-only artifacts. This work proposes BVOLT: a modular and generalizable software designed to simulate, model, and control microgrid components. The project's primary goal is to create a flexible, easy-to-use interface between MGDTs and end-users to enhance testing and interactions. As a testbed, a 5-cell lithium battery bank Digital Twin is proposed as a MGDT component to validate Microgrid Digital Twins creation and validation using physics-based and data-driven models. BVOLT is expected to be a front in MGDT components development as a useful link between researchers and physical apparels.

Keywords: Microgrids, Renewable Energy Resources, Digital Twins

LIST OF FIGURES

Figure 1 – Diagram representing LabInovação's microgrid components, electric connections and modbus protocol communication links. 26
Figure 2 – The laboratory microgrid whole software structure at the time being, from the entry point to the user presentation. . 27
Figure 3 – Integrated architecture combining SOC estimation, Digital Twin modeling, and the BVOLT platform for remote monitoring and control. 46
Figure 4 – The project work plan chronogram for the remaining of the year of 2026. 47

LIST OF TABLES

Table 1 – Solar and hydrogen plant physical entities. 24
Table 2 – Solar thermal plant physical entities. 25
Table 3 – Comparison of physics-based and data-driven Digital Twin models 44

LIST OF ABBREVIATIONS AND ACRONYMS

ABNT	Brazilian Association of Technical Standards
abnTeX	Absurd Standards for TeX

CONTENTS

I RESEARCH 20

1 INTRODUCTION 21

2 MATERIALS AND METHODS 24

3 LITERATURE REVIEW 28

3.1 MICROGRID ASSETS 29

3.1.1 Renewable Energy Resources 29

3.1.2 Energy Storage Systems 30

3.1.3 Power Electronics and Grid Infrastructure 31

3.2 THE VIRTUAL SYSTEM, HIGH-FIDELITY MODELS AND FUNCTIONS 31

3.2.1 Modeling Methodologies 32

3.2.2 Core Virtual Functions and Services 32

3.3 DATA EXCHANGE AND ENABLING TECHNOLOGIES . . . 34

3.3.1 Data Types and Acquisition 34

3.3.2 Infrastructure and Communication 34

3.4 CHALLENGES 35

3.4.1 Data Management and Processing Hurdles 36

3.4.2 Modeling and Simulation Complexity 36

3.4.3 Connectivity, Interoperability and Standardization 37

3.4.4 Security, Privacy and Ownership Concerns 39

3.4.5 Economic and Implementation Costs 39

3.4.6 Microgrid-specific and Foundational Challenges 40

II PROPOSAL 42

4 BVOLT 43

5 OBJECTIVES 45

6 CHRONOGRAM 47

7 PERIODICALS AND SCIENTIFIC EVENTS 48
7.1 PERIODICALS (JOURNALS) 48
7.2 SCIENTIFIC EVENTS (CONFERENCES) 49

REFERENCES 52

Part I

Research

1 INTRODUCTION

Due to high greenhouse gases (GHGs) emissions over the atmosphere, specially methane (CH_4), nitrous oxide (N_2O) and carbon dioxide (CO_2)—while the latter emissions mainly come from fossil fuels and industrial activity (Allen *et al.*, 2018)—, the global surface temperature has risen to dangerous and unprecedented levels. High global surface temperature leads to droughts, floods and sea level rise, ice sheet melt, biodiversity loss, and extended heatwaves, leading to irreversible changes in nature and human life as a whole. With a worrisome climate perspective on the horizon, renewable energies have been seen as the promise for a sustainable world. In this line, international effort has been put in decarbonizing energy consumption activities and enhancing energy distribution management systems.

However, renewable energy resources (RERs) are intermittent, and, coupled with variable load profiles, lead to energy imbalances and power quality issues in power grids (Shahzad *et al.*, 2022). Distributed energy generation such as microgrids can address these problems and rely on small-scale onsite generation of energy resources (Joshal; Gupta, 2023). A microgrid (MG) is a low-voltage power system network comprised of distributed energy resources, energy storage systems, and power converters (Jamal; Tan; Pasupuleti, 2021; Mishra; Shankar, 2025; Kumari *et al.*, 2023)—it is an electrical system for generating, storing and consuming energy in uninterrupted fashion. Microgrids can vary in size and topology of its components. They can also be connected to a central power grid through a point of common coupling (PCC) or not and, hence, be called, respectively, power connected (*on-grid*) or islanded (*off-grid*) microgrids. The integration of microgrids in centralized power connected networks is complex, and demands simulation models—digital twins (DTs) are capable of resolving this integration with an accurate digital entity of physical components (Kumari *et al.*, 2023).

Born within the aerospace industry, the digital twin technology aims to mirror a physical device or process. For their creation and usage, it is imperative to have a bidirectional data flow between the twins and proper models, which

can be physical-based, data driven or a combination of both approaches. Inside the context of microgrids, microgrid digital twins (MGDTs) operate to enhance energy systems efficiency, remaining useful life, cost, service quality and asset management, while applied to health monitoring, forecasting, fault detection, control management, decision-making, and cyber-security (Kumari *et al.*, 2023). However, the development of simulation and control models for microgrids is time-costly and error-prone Software implementations to simulate, model and control microgrids, and their respective digital twins, is nontrivial.

Developing a modular generalizable environment for the creation and adaptation would a) speed MGDT deployment, and b) lower the probability of errors and rework during the plant design and operation. With these hypothesis in mind, this work relies in the creation of a Microgrid Digital Twin component as its proof of concept (PoC)—a battery bank composed of 5 UNICOBA UPLFP48 lithium batteries. In this scenario, INCT-CAPE LabInovação's microgrid is posed as a good test-bed for this software piece.

The National Institute of Science and Technology (INCT) is a country-wide initiative of several institutes for research; formation of human resources; knowledge, technology and innovation transfer to the society; and internationalization (Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq), 2025). Their objective is to improve diverse fields of agricultural, ecology, energy, exacts, social, nanotechnology, health and engineering sciences. The INCT control and automation of energy processes institute (INCT-CAPE) is composed of a myriad of laboratories, such as LabInovação— an on-grid auto-sufficient microgrid, described in Chapter 2. With all being said, the project called BVOLT works as a PoC for MGDTs as it works a Microgrid Digital Twin storage component—therefore, it is focused entirely on the study and development of microgrid digital twins.

Apart from the introduction, this work plan explains BVOLT by: explaining LabInovação's materials and methods at Chapter 2; presenting a literature review tangent to microgrid digital twins at Chapter 3; and, at last, proposing a five-cell battery bank software to be deployed at the end of the project at Part II, which is composed of 4 chapters: 1) the dissertation main hypothesis and the proposed piece of software to solve it; 2) the general and specific objectives of

the research; 3) the project chronogram; and 4) periodicals and scientific events in which the author could provide research submissions and presentations.

2 MATERIALS AND METHODS

This section establishes all of LabInovação's microgrid physical and digital components, and describes how their connections are made. The laboratory physical entity components are divided in 2 plants, described at Tables 1 and 2. This project focus exclusively at the hydrogen and solar PV plant, as the solar thermal plant is still in design process and, thus, will not be considered for this work project.

The battery bank, PV solar arrays, electric generator emulator and wind generator emulator are all power connected to the main, and first, inverter—which is connected to the power grid. The child, and second, inverter is power connected to the other by a father-child scheme, and is connected to another set of PV solar arrays. The second inverter is also responsible for managing generation, storage and consume from the plant hydrogen-related equipment. The solar and hydrogen plant has a MODBUS network that connects logically the main computer and the main inverter. All electrical connections can be seen at Figure 1.

The main computer runs a source code entirely written using the Python programming language (Python Software Foundation, 2025), which excels in the ease of building complex data structures and handling data science func-

Table 1 – Solar and hydrogen plant physical entities.

Solar and hydrogen plant
10+11 photovoltaic solar arrays
Electric generator emulator
Battery bank of 5 UNICOBA UPLFP48 lithium batteries
2 DEYE SUN-12K-SG04LP3-EU three-phase inverters
Proton Exchange Membrane Fuel Cell (PEMFC)
Proton Exchange Membrane (PEM) electrolyzer
Metal hydrate storage tank
Wind generator emulator

Table 2 – Solar thermal plant physical entities.

Solar thermal plant
Solar thermal collectors
Solar thermal storage tanks with electric heaters (boilers)
Pumps
Heat exchangers
Valves
Thermal sink

tionalities without being verbose—the physical laboratory entities are described through the Object-Oriented Programming (OOP) paradigm. The source code is run inside a UV virtual environment (Astral, 2024). Data exchange between devices is stored as time-series data inside an InfluxDB database (InfluxData Inc., 2025). All obtained database information is shown at a Grafana dashboard (Grafana Labs, 2025), which has intuitive information handling operations and work as a plug-and-play visualization environment for the whole project ecosystem. The SOLID methodology is used in order to build a clean code and a clean architecture (Martin, 2017), objectifying software maintainability, flexibility, and scalability. The SOLID methodology represents a piece of software that follows the Single responsibility, Open/closed, Liskov substitution, Interface segregation, and Dependency inversion principles; for more information about each principle, please relate to (Martin, 2008).

The main computer running the source code is responsible for executing all software services, which are:

- Software communication interface. A software interface to connect the physical devices, communicating with the inverter by a Modbus connection. Its related registers table is used in order to access the proper functional addresses, which are read and writen through the code execution.

- Control. A Model Predictive Control (MPC) that regulates the battery bank state of charge (SOC) considering its phenomenological model for predicting its charging/discharging power according to a given demand curve.

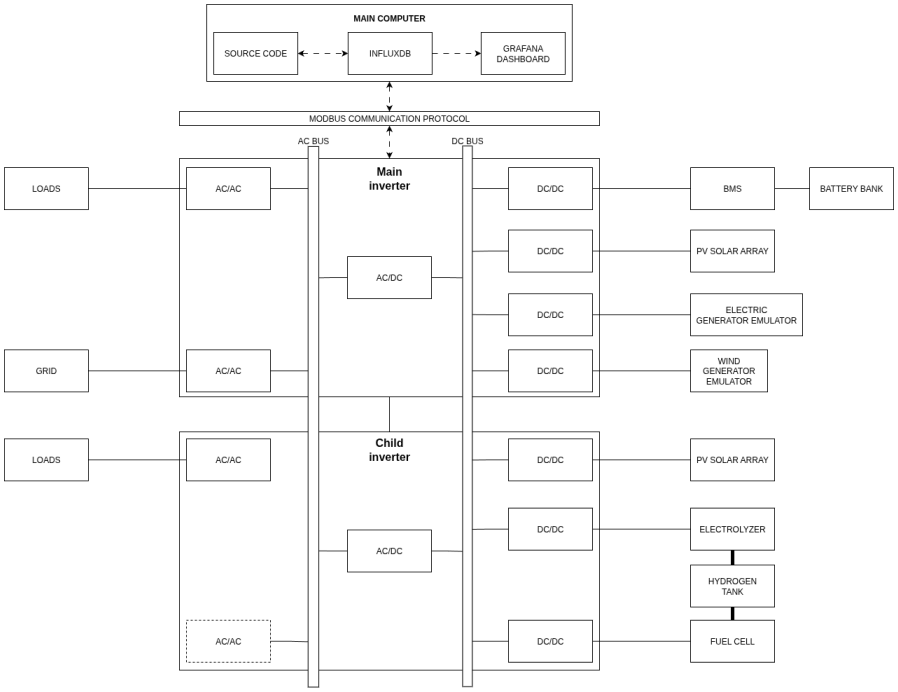

Figure 1 – Diagram representing LabInovação's microgrid components, electric connections and modbus protocol communication links.

- Weather forecasting. A software connection to read the time of the day and weather conditions.
- Data management. An InfluxDB database connection through a localhost server—for storing and making accessible process variables data.

This repository follows a clean, installable Python `src/` layout driven by `pyproject.toml`, with a thin `main.py` entry point and a core `microgrid_integration` package that enforces clear separation of concerns. Configuration and runtime settings are centralized via environment-oriented modules and static Modbus register tables, while a dedicated device abstraction layer encapsulates hardware concepts through base interfaces and concrete implementations (e.g., inverters). Protocol adaptation is handled by a Modbus loader that maps register definitions into queryable runtime structures. Persistence concerns are isolated in a storage adapter targeting InfluxDB, enabling decoupled time-series ingestion. Domain and service logic aggregate and normalize raw telemetry

into higher-level metrics, which then feed an MPC and optimization subsystem containing microgrid models, parameter definitions, reference trajectories, convex-based controllers, and forecasting utilities. A lightweight presentation layer provides tabular formatting for outputs, and the overall structure is complemented by auxiliary `data/` inputs for control experiments, `references/` with vendor and protocol documentation, and a minimal `tests/` footprint, forming a coherent pipeline from field acquisition to analytics and control. The `microgrid-integration` structure can be seen at Figure 2.

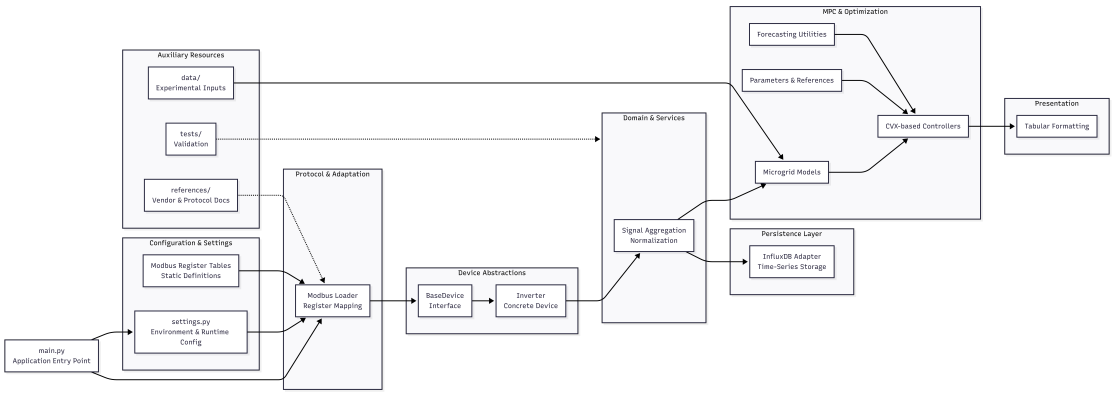

Figure 2 – The laboratory microgrid whole software structure at the time being, from the entry point to the user presentation.

3 LITERATURE REVIEW

The Fourth Industrial Revolution bears Industry 4.0 as the next step for autonomous and intelligent technology systems. Advancements in the fields of Internet of Things (IoT), cyber-physical systems (CPSs), artifical intelligence (AI), and data science analytics are the drivers for more connected automation processes (Bazmohammadi *et al.*, 2022). This new industrial concept focus at state-of-the-art computational paradigms, such as remote monitoring, web-based interfaces, cloud computing, cybersecurity, IoT devices and Digital Twins (DTs) (Folgado *et al.*, 2024). These mechanisms are employed as tools for solving control, automation and optimization problems in the fields of process efficiency, production management, maintenance planning, sustainability and energy (Folgado *et al.*, 2024)—the latter being under profound transformation, driven by the dual imperatives of decarbonization and decentralization.

In this context, the proliferation of Distributed Energy Resources (DERs)—such as solar photovoltaics (PV), wind turbines, and battery Energy Storage Systems (ESSs)—is fundamentally altering the architecture and operational dynamics of traditional energy systems, such as power grids (Das *et al.*, 2024). While these technologies are critical for a sustainable energy future, their inherent intermittency and the power-electronic-interfaced nature of their grid connection introduce significant challenges related to stability, control, and reliability (Das *et al.*, 2024). Microgrids have emerged as a key solution to manage these complexities, enhance local resilience, and facilitate DER integration—albeit with the cost of complex interplay of diverse assets, variable generation, and dynamic loads, which requires advanced monitoring, control, and optimization capabilities that often exceed the capacity of conventional Supervisory Control and Data Acquisition (SCADA) systems.

Concomitantly, DTs have emerged as a key technology—high fidelity, virtual representations of physical assets, systems, or processes which communicate by a seamless, bidirectional link between the physical and virtual worlds, enabling comprehensive analysis, simulation, and prediction of the physical twin behavior across its entire lifecycle (Kumari *et al.*, 2023). Therefore, a Microgrid

Digital Twin (MGDT) applies this concept to the entire microgrid ecosystem, creating a virtual replica that mirrors the real-time state and dynamic behavior of all its components (Bazmohammadi *et al.*, 2022), including generators, storage, loads, and the underlying communication and control networks. The utilization of Microgrid Digital Twin (MGDT) technology represents a revolutionary paradigm shift in the energy sector, offering a comprehensive virtual modeling platform that significantly enhances the operation, integration, and management of modern electrical systems (Feng *et al.*, 2024). This extensive literature review delves into the complex components, enabling technologies, and functional applications that constitute the MGDT architecture, drawing upon the comprehensive insights provided in the source material.

The fundamental value proposition of a MGDT lies in its ability to serve as a risk-free environment for testing, validation, and optimization (Bazmohammadi *et al.*, 2022). By running simulations on the virtual model, operators can explore "what-if" scenarios, predict the impact of control actions before implementation, diagnose faults, and anticipate future performance degradation without jeopardizing the stability of the physical microgrid. This review synthesizes the current state of research and development in MGDT technology.

3.1 MICROGRID ASSETS

The physical system, or Physical Twin, is one of the foundations of the DT, encompassing the real-world assets that operate and generate data (Liu *et al.*, 2023). The MGDT is established by creating a Digital Twin of each section of the microgrid (Bazmohammadi *et al.*, 2022). Power systems are considered complex human-made systems, facing constant challenges related to the connection of Renewable Energy Resources (RERs), the integration of power electronics equipment, and energy management—all of which necessitate regular monitoring (Kumari *et al.*, 2023).

3.1.1 Renewable Energy Resources

DT applications are crucial for the smooth incorporation of distributed energy resources into power grids. Digital Twins are applied to Photovoltaic (PV)

installations for assessment, monitoring, and control (Das *et al.*, 2024). DT models of PV components can be used to predict Remaining Useful Life (RUL) of components (Das *et al.*, 2024), fault detection and Maximum Power Point Tracking (MPPT) (Angelova *et al.*, 2024). DTs are also used to forecast the Remaining Useful Life (RUL) of power converters in variable-speed wind turbines, often utilizing credible physics-based models (Das *et al.*, 2024; Kumari *et al.*, 2023)—DT implementation in wind farms provides detailed insights into topography and climatic conditions (Sahoo *et al.*, 2025). For biogas, predictive models utilizing machine learning and deep learning (such as RNN-based models) help optimize waste selection and forecast biogas production (Kumari *et al.*, 2023). Tangent to hydrogen, DT technology enhances the operation and integration of hydrogen systems into broader energy networks—DTs are established to obtain full knowledge of the Proton Exchange Membrane Fuel Cell (PEMFC) real-time state through 3D virtualization-multi-physics DTs (Feng *et al.*, 2024), and DT models also support model-based design methodologies and reduce performance degradation in fuel cells (Feng *et al.*, 2024).

3.1.2 Energy Storage Systems

Energy Storage Systems are vital for microgrids, storing excess energy (e.g., during off-peak hours) and helping maintain energy balance (Kumari *et al.*, 2023). ESSs are contemplated by: 1) batteries and electric vehicles; and 2) hybrid and seasonal storage. With what is related to batteries, their internal status is hard to measure and traditional estimation methods for State of Charge (SOC) and State of Health (SOH) are challenging with degrading batteries (Kumari *et al.*, 2023)— a perfect fit for Digital Twins. The DT framework for EV battery management consists of the on-vehicle Battery Management Systems (BMS)—the physical system—, the Battery Digital Model (virtual system), and the cloud-based interconnection—the data link (Eaty; Bagade, 2023). Computationally intensive SOH prediction is performed remotely in the cloud to reduce on-vehicle compute requirements and overall costs (Eaty; Bagade, 2023). Real-time SOC estimation, critical for range prediction, is computed on the on-vehicle BMS (Eaty; Bagade, 2023). Advanced methods like Kalman filters are used for

SOC estimation, often combining Coulomb counting and Open Circuit Voltage (OCV) techniques for precise estimations (Eaty; Bagade, 2023). Continual learning approaches are also proposed for incremental fine-tuning of deep learning networks used for SOH prediction (Eaty; Bagade, 2023). Concomitantly, DTs are essential for hybrid ESSs, which integrate multiple storage technologies (like batteries and hydrogen) to mitigate individual disadvantages (Oliveira Luna *et al.*, 2024). DTs facilitate the design and operation of long-term energy storage systems, especially when integrating seasonal storage capacity, such as hydrogen production via water electrolysis. (Oliveira Luna *et al.*, 2024; Feng *et al.*, 2024).

3.1.3 Power Electronics and Grid Infrastructure

Power converters are the third most crucial part of a MG, establishing connections among various components (Kumari *et al.*, 2023). DT methodologies have been explored for the monitoring and control of DC-DC converters (Lei *et al.*, 2023). A DT DC-DC converter can be designed and implemented using a hybrid twin modeling method (Lei *et al.*, 2023). This approach can dynamically track the physical system even when the system model varies and can ensure a functioning system by detecting and replacing a failed physical controller (Lei *et al.*, 2023). The MGDT provides a high-fidelity simulation platform that enables the efficient determination of the appropriate size and capacity of system equipment, including generators, transformers, lines, switches, converters, and ESSs (Bazmohammadi *et al.*, 2022).

3.2 THE VIRTUAL SYSTEM, HIGH-FIDELITY MODELS AND FUNCTIONS

The Virtual Model, or Virtual Twin, is the computational copy that characterizes and describes the physical entity from multiple temporal and spatial dimensions (Liu *et al.*, 2023). The model is paramount to the DT and a prerequisite for successful applications (Tao *et al.*, 2022).

3.2.1 Modeling Methodologies

DT creation requires the integration of high-fidelity models with multi-source data such as sensor data and historical information (Bazmohammadi *et al.*, 2022). The models can be severed into 3 main categories: 1) physics-based models; 2) data-driven models; and 3) hybrid models.

The physics-based modeling approach uses mathematical and theoretical concepts (e.g., differential equations or transfer functions) to model the physical behavior (Machado *et al.*, 2023a). It is necessary to reflect the physical properties, characteristics, and constraints of the entity (Tao *et al.*, 2022). Examples include the use of Partial Differential Equations (PDE) for phenomenological modeling of solar collectors (Machado *et al.*, 2023a).

Data-driven models rely on algorithms and statistical models, such as the ones employed with AI techniques, to learn from data alone (Rasheed; San; Kvamsdal, 2020). These models exploit tacit knowledge and historical data to make the DT model smarter (Tao *et al.*, 2022). DTs investigate the synergy between DT and AI to advance predictive analytics and optimize system performance (Das *et al.*, 2024). Examples include using Adaptive Neuro-Fuzzy Inference Systems (ANFIS), which are soft computing techniques preferred when data is incomplete, noisy, or when phenomenological models are complex and computationally expensive (Machado *et al.*, 2023a,b).

Hybrid models are considered a promising solution that combines the interpretability and robust foundation of physics-based models with the accuracy and efficiency of data-driven models (Bazmohammadi *et al.*, 2022; Rasheed; San; Kvamsdal, 2020). The objective of this hybridization is to remove the shortfalls of relying purely on either physics-based or data-driven models (Rasheed; San; Kvamsdal, 2020).

3.2.2 Core Virtual Functions and Services

DT models are designed to provide real-time, efficient, and intelligent services (Liu *et al.*, 2023). The virtual environment is where techniques like simulation, modeling, and evaluation take place (Jones *et al.*, 2020).

DTs significantly enhance forecasting capabilities, which is one of the most

significant tasks of MGs (Das *et al.*, 2024; Bazmohammadi *et al.*, 2022). Accurate prediction of RERs available power is crucial for system reliability and ensures maximum resource utilization (Bazmohammadi *et al.*, 2022; Kumari *et al.*, 2023). DTs also predict system evolution and project future conditions (Bazmohammadi *et al.*, 2022). Predictive capability, such as Long Short Term Memory (LSTM) networks, is instrumental in creating future scenarios when direct observation data is unavailable (Rasheed; San; Kvamsdal, 2020).

DTs facilitate optimal control and coordination of interconnected energy assets—as DTs being essentially dynamic simulation engines that provides near-real-time visualization for power systems (Das *et al.*, 2024). Model Predictive Control (MPC) algorithms, which work with a prediction horizon, are commonly used for optimal energy management in nanogrids (NGs) and smart houses (Oliveira Luna *et al.*, 2024). MPC formulation is often achieved as a Mixed-Integer Quadratic Programming (MIQP) optimization problem, which avoids the complexities of nonlinear approaches (Oliveira Luna *et al.*, 2024). In MPC, the cost function can incorporate a QoE metric to balance economic costs (minimizing energy bills) with user comfort and satisfaction, particularly concerning demand management actions (Oliveira Luna *et al.*, 2024).

MGDTs provide a holistic, up-to-date dynamic view of the physical system's operating conditions and assets, thereby improving Operational Situational Awareness (OSA) and resilience (Bazmohammadi *et al.*, 2022; Das *et al.*, 2024). Resilience is the system's ability to anticipate, absorb, recover from, and adapt after high-impact low-priority (HILP) events (Bazmohammadi *et al.*, 2022). MGDTs enhance resilience by providing a high-fidelity platform to assess the effects of preventive, corrective, and restorative actions against threats, including natural hazards (e.g., hurricanes, earthquakes) and human-induced hazards (e.g., cyber-attacks, malicious attacks) (Bazmohammadi *et al.*, 2022).

DTs are crucial for cybersecurity applications, enabling the real-time modeling of cyber and physical layers (Danilczyk; Sun; He, 2019). An Intelligent Digital Twin framework, such as ANGEL (Automatic Network Guardian for ELectrical systems), is designed to create a learning and self-healing system necessary to defend the power grid, enhancing real-time response and damage mitigation (Danilczyk; Sun; He, 2019). DTs can serve as a cyber-physical reference

model for real-time attack localization. DTs enhance fault diagnosis by comparing DT output with physical output in real time (Angelova *et al.*, 2024; Kumari *et al.*, 2023).

3.3 DATA EXCHANGE AND ENABLING TECHNOLOGIES

The connection, often termed the Digital Thread, is the dynamic, data-driven support structure that facilitates the seamless, bidirectional exchange of information (Singh *et al.*, 2021; West; Blackburn, 2017). This continuous link is what differentiates a DT from traditional multi-physics simulation and modeling approaches (Bazmohammadi *et al.*, 2022; Danilczyk; Sun; He, 2019).

3.3.1 Data Types and Acquisition

The DT requires well-developed infrastructures for collecting high-quality data. Data are collected through sensors, IoT-enabled devices and smart meters, and are divided into 4 types: 1) physical data; 2) virtual data; 3) knowledge data; and 4) derivative data (Liu *et al.*, 2023).

- Physical data. Raw data retrieved from sensors;
- Virtual data. Data generated from the model itself, e.g. simulation results;
- Knowledge data. Technical information, historical operational data, and maintenance history; and
- Derivative data. Processed and analytical information, such as SOC and SOH estimations.

3.3.2 Infrastructure and Communication

Efficient data handling is paramount, especially given the vast increase in data volume from sophisticated sensors connected to the grid—hence, it is worth mentioning computational infrastructure frameworks; communication and interoperability technologies; and standardization.

Cloud computing and edge computing (edge intelligence) compose the

computation infrastructure paradigms used to host data from or by the digital twin and to properly analyze and process them. Cloud-based computing serves to overcome the issue of processing the immense amount of sensor data generated by the grids; it sends the obtained data to dedicated remote servers to pre-process, filter and store what is collected from the twins. Edge computing implements artificial intelligence at the network edge—at the physical facility or plant related to the microgrid—to perform data analysis *in loco*; this reduces latency of transmission and high bandwidth requirements, while also enhancing data privacy.

Notwithstanding, related to communication, the connection must always be bidirectional, and it is vital for both physical and digital twin to maintain consistent data throughput. To reduce communication latency, high-speed networks, such as MQTT-UA, 5G or LoRAWAN, are necessary (Sahoo *et al.*, 2025; Smith; Robinson; Agalgaonkar, 2021).

Albeit the digital twin is a trending topic, there is a lack of standardization among its data models and computational infrastructure, and communication protocols, which is posed as a key challenge in microgrid implementation, hindering seamless interoperability and scalability. International standards organizations, such as the IEC Technical Committee 57 (TC 57), focus on developing standards for power systems control equipment and associated information exchange (Gillerman; Falk; Mackiewicz, 2005).

3.4 CHALLENGES

The challenges of MGDT implementation include dealing data management; modeling complexity; connectivity and standardization; security and privacy; and cost. The development and implementation of Digital Twins for Microgrids (MGDTs), or DTs applied to power systems and smart grids, faces a multitude of technical, organizational, and economic hurdles (Das *et al.*, 2024; Singh *et al.*, 2021). These challenges stem from the inherent complexity of power systems, the high demands of real-time data handling, and the novelty of the DT technology itself (Das *et al.*, 2024; Singh *et al.*, 2021).

3.4.1 Data Management and Processing Hurdles

Effective MGDT operation relies on high-quality, continuous data streams from sophisticated sensors and IoT devices deployed throughout the system (Danilczyk; Sun; He, 2019; Das *et al.*, 2024; Kumari *et al.*, 2023).

- Data volume, velocity and quality. DT application in power systems requires a highly advanced virtual environment capable of handling a large amount of data—it is difficult and requires significant computer power to manage, process and analyze data in real time (Das *et al.*, 2024).
- Data acquisition and storage. Data collected from real scenarios are either locally decentralized or centrally stored in a cloud (Das *et al.*, 2024). Data management, storage, rapid response, computing needs, and scalability are complexities that necessitate efficient power systems management (Das *et al.*, 2024).
- Poor data quality. Poor data quality and missing information reduce the possibility of accurate DT execution, leading to inadequate results (Das *et al.*, 2024; Segovia; Garcia-Alfaro, 2022).
- Data synchronization and time issues. Data must be synchronized between both twins to ensure accuracy and reliability in smart microgrid applications; divergence can lead to performance degradation errors (Sahoo *et al.*, 2025)—the issue of time skewness in data acquisition, caused by different reporting rates of sensors, complicates the overall problem of achieving a comprehensive MGDT (Kumari *et al.*, 2023; Rasheed; San; Kvamsdal, 2020).

3.4.2 Modeling and Simulation Complexity

Creating a virtual representation of complex, dynamic power systems is a difficult task, requiring advanced modeling strategies and intense computational resources (Das *et al.*, 2024).

- Replication complexity. Power systems are highly complex, characterized by advanced interactions and behaviors among interrelated components

like generators, transmission lines, substations, and distribution networks (Das *et al.*, 2024). Developing a precise replication of a physical asset while accurately adjusting its physical parameters can be a difficult task (Das *et al.*, 2024).

- Integrating RERs. MGs integrate RERs, which are characterized by their significant intermittency and dependence on weather conditions (Das *et al.*, 2024). The development of the virtual representation of these dispersed energy resources is more difficult (Das *et al.*, 2024)—e.g. for solar photovoltaic (PV) systems, the dependence on environmental conditions makes it challenging to recreate different scenarios for simulation using a DT (Angelova *et al.*, 2024).

- Computational Demand and Efficiency. The computational requirements for modeling and simulation are very intensive and need to be very efficient (Das *et al.*, 2024). A key challenge is efficiently balancing the trade-off between the accuracy of predictions by digital twins and optimizing the computational complexity required for various types of models and data (Feng *et al.*, 2024).

- Model Interpretability and Validation. It is essential to know and explain the process by which these models generate their predictions or decisions, especially in important applications where human operators must have trust in and understand the logic behind the model's behavior (Das *et al.*, 2024; Rasheed; San; Kvamsdal, 2020). Furthermore, modeling complexity is infeasible to construct in one go, requiring complex processes of model assembly, fusion, verification, modification, and management, which are not yet fully addressed in current research (Tao *et al.*, 2022).

3.4.3 Connectivity, Interoperability and Standardization

Seamless, real-time communication between the physical microgrid and its virtual twin is crucial, yet constrained by technological limitations and a lack of universal protocols (Das *et al.*, 2024).

- Connectivity. Establishing a connection between the physical system and its virtual counterpart is crucial for effective DT implementation, allowing for real-time monitoring and remote operations (Das *et al.*, 2024). However, problems remain with connectivity, such as software mistakes and power outages (Das *et al.*, 2024). Effective implementation is limited by the latency in data collecting of physical devices, the quickness of communication in reaction to real-time events, and the compatibility of interaction speed across various subsystems of DTs (Das *et al.*, 2024; Rasheed; San; Kvamsdal, 2020).

- Interoperability. DTs must combine data from several systems and devices, often using different communication protocols and standards (Das *et al.*, 2024). The ability to collect, aggregate, and exchange data with different components and suppliers presents many interoperability challenges, especially under real-time constraints or proprietary interfaces (Segovia; Garcia-Alfaro, 2022). Furthermore, the evolution and adoption of applications are limited by the need for interoperability standards that allow different assets and DTs to work together within a large architecture (Segovia; Garcia-Alfaro, 2022; Rasheed; San; Kvamsdal, 2020).

- Standardization. The diversity of potential DTs for various industries and purposes presents a significant challenge for standardization in DT technology (Das *et al.*, 2024). There must be a standardized method from the initial design phase to the simulation of a DT (Das *et al.*, 2024; Segovia; Garcia-Alfaro, 2022). For microgrids, the lack of standardization in communication protocols and data models hinders integration and scalability, especially when dealing with multiple legacy systems with incompatible data formats (Sahoo *et al.*, 2025). In the context of Multi-Agent Systems (MAS) used in microgrids, standards are crucial for the agent communication language, protocol, and ontology to ensure interoperability (Smith; Robinson; Agalgaonkar, 2021).

3.4.4 Security, Privacy and Ownership Concerns

Due to their reliance on collecting and exchanging sensitive data and their network connectivity, DTs pose significant security and ethical challenges (Das *et al.*, 2024; Javaid; Haleem; Suman, 2023).

- Cybersecurity Vulnerability. DTs are highly vulnerable to cyber dangers and attacks due to their connection to networks and systems (Das *et al.*, 2024; Kumari *et al.*, 2023). The integration of thousands of sensors and measurement devices increases the vulnerability to False Data Injection Attacks (FDIA) (Das *et al.*, 2024). The bidirectional communication required in DTs means that security must be monitored for both the physical system and the virtual layer (Kumari *et al.*, 2023).
- Privacy and Data Protection. Protecting privacy and security is crucial as DT applications rely on sensitive data from many sources (Das *et al.*, 2024). Data protection and jurisdiction issues can arise, particularly in cross-border data exchanges, necessitating legal compliance and proper governance structures (Sahoo *et al.*, 2025). Protecting proprietary data from exposure, such as forecasting models and consumption patterns used in advanced analytics, is also critical (Sahoo *et al.*, 2025).
- Data Ownership. Determining data ownership is complicated due to the involvement of different stakeholders who participated in the generation, processing, value addition, or analysis of data (Rasheed; San; Kvamsdal, 2020). An unanswered question exists regarding whether an individual owns the virtual twin and associated data of a product they purchase, especially when data is needed for forensic or legal purposes (Jones *et al.*, 2020).

3.4.5 Economic and Implementation Costs

The realization of DT technology, especially in complex systems like microgrids, demands substantial financial investment and expertise, often with an unclear Return on Investment (ROI) (Javaid; Haleem; Suman, 2023; Sahoo *et*

al., 2025).

- High Initial Investment. DT incurs significant costs in infrastructure development, mainly due to substantial investments in hardware, software, and networking infrastructure (Das *et al.*, 2024). High initial investment is required for sensors, IoT devices, and integrating complex systems, which smaller organizations may find difficult to justify if the ROI is not immediately clear (Sahoo *et al.*, 2025).

- Cost and time quantification. The process of developing high-fidelity computer models is time-consuming and labor-intensive, and requires huge computational power—hence, the DT being an expensive investment (Singh *et al.*, 2021).

- Maintenance costs. Maintaining and ensuring the cybersecurity systems, as well as periodically upgrade them, are expensive operations (Das *et al.*, 2024).

3.4.6 Microgrid-specific and Foundational Challenges

DT technology, being relatively nascent, struggles with fundamental implementation issues when applied to complex systems like MGDTs (Das *et al.*, 2024; Singh *et al.*, 2021).

- System complexity in microgrids. Major challenges in implementing microgrids generally include optimal power flow, stability, protection, resynchronization after fault, RER integration, and harmonics (Kumari *et al.*, 2023). The MGDT concept must address the complexities related to grid integration, security, and redundancy (Kumari *et al.*, 2023).

- Scalability. DT systems produce large amounts of data, necessitating power computers for real-time processing and analysis—which causes issues when applying DT in complicated and extensive power systems (Das *et al.*, 2024), such as microgrids.

- Novelty and lack of expertise. DT is still and emerging technology, lead-

ing to a lack of clean understanding about the value it brings, insufficient case studies, and lack of know-how regarding the technical and practical knowledge required for implementation (Singh *et al.*, 2021).

- Lifecycle mismatching. For assets with lifecycles, e.g. grid infrastructure, the lifespan of the physical product is often longer than the validity of the software and data storage methods used for the DT, raising concerns about long-term maintenance (Singh *et al.*, 2021).

Part II

Proposal

4 BVOLT

This chapter presents the Digital Twin (DT) architecture to be developed and its proof of concept considering a lithium battery bank of the LabInovação microgrid. The DT will consist of two complementary modeling approaches: a physics-based and a data-driven models. Both to-be-decided models will be thoroughly compared against each other to verify which can more accurately predict the 5-battery bank SOC according to its injected and ejected power. The digital-physical interaction is established through Modbus communication, remote data storage via InfluxDB, dashboard visualization via Grafana, and integrated deployment inside the BVOLT software. The InfluxDB and Grafana dashboard will both be hosted in a dedicated server to be accessible remotely.

BVOLT is a modular software deployment framework that combines real-time streaming and modeling approaches such as physics-based and data-driven. This work considers BVOLT as a proof of concept in LabInovação's MGDT. The use of Modbus communication, InfluxDB data storage, and integrated physics-based and data-driven models enables robust predictive control and long-term state of charge forecasting of lithium battery systems in microgrid environments.

The Digital Twin will be composed of three main components: 1) the physical battery bank and BMS; 2) the cyber infrastructure for data streaming, storage and synchronization; and 3) the virtual twin hosting physics-based and data-driven models.

The physics-based representation focuses on estimating the battery state of charge using a state-space formulation described through differential equations that capture the underlying physical behavior of the battery. The model parameters are obtained experimentally. Real-time sensor data is streamed through Modbus and stored in the database—where it is used to continuously update and correct the model states. The data-driven model DT also targets the SOC estimation, but does so by learning the relationship between measurements and state of charge directly from the data gathered through the 5-battery bank lifecycle operation. It uses historical and real-time multivariate time-series

data to identify non-linear dependencies associated with the SOC alteration through the day. This approach is effective when dealing with complex, difficult-to-describe dynamics, measurement noise, or operational conditions not fully captured by physics-based models. As more data becomes available, the model adapts to new usage patterns—which enables accurate SOC forecasting over different time horizons. All features are retrieved from the same database infrastructure used in the physics-based model to ensure consistent and comparable data management.

Table 3 – Comparison of physics-based and data-driven Digital Twin models

Approach	**Main Purpose**	**Strengths**
Physics-Based Model	Operational analysis and physically interpretable predictions	Captures underlying electrochemical behaviour; constraint and safety awareness; robustness under well-defined conditions
Data-Driven Model	Forecasting and identifying long-term performance trends	Learns complex patterns; adaptable to new data; effective for SOC estimation
Hybrid Model	Integrates physics insights with data-driven adaptability	Improved lifetime management; enhanced accuracy; complementary strengths

The Modbus data link constitutes the core of the data exchange, providing low-latency communication between the physical and virtual twins. The BMS has registers including battery voltage, current, temperature, SoC, alarms, and cycle number, which are periodically polled through the Python `minimalmodbus` library. These values are stored in time-series format using InfluxDB, enabling both real-time control applications and long-term predictive modeling.

5 OBJECTIVES

The overall goal of this work is to create a 5-cell lithium battery bank DT to test case MGDTs components and to facilitate further Microgrid Digital Twins creation for other researchers worldwide. The specific objectives reflect validating physics-based and data-driven models for the 5-cell battery digital twin.

- Make the existing InfluxDB database remotely accessible to enable online data curation and DT operations;
- Make the existing Grafana dashboard cloud-enabled for remote development and worldwide usage by other research teams;
- Build a robust software interface with plug-and-play functionalities for creating a 5-battery bank DT as a MGDT component;
- Estimate the 5-cell lithium battery bank SOC using a physics-based model;
- Estimate the 5-cell lithium battery bank SOC using a data-driven model;
- Head-to-head comparison defining and running comparative experiments between the physics-based and data-driven models across the same test scenarios;
- Integrate and deploy integration chosen estimators into BVOLT's DT stack with data ingestion from Modbus to InfluxDB, produce Grafana dashboard for live SOC curves, and ensure inference can run on the target device (edge or server); and
- Publish code, datasets and results in GitHub with reproducible notebooks that recreate main figure, and submission of an article describing the done comparative study

Figure 3 illustrates the integrated architecture for state-of-charge (SOC) estimation, digital-twin integration, and remote access implemented in the BVOLT platform. The physical layer is composed of a five-cell lithium battery bank instru-

mented by BMS sensors, whose measurements are acquired via Modbus protocol by a dedicated polling worker that isolates hardware access from higher-level services. The polling worker writes telemetry to an InfluxDB instance, which acts simultaneously as a time-series database and a state cache, while also reading command updates generated upstream. From this centralized state layer, SOC estimation is performed in parallel using a model-based estimator and a data-driven estimator, enabling methodological comparison under identical operating conditions. The outputs of both estimators are fed into a benchmarking module, whose results drive a digital twin environment responsible for state twinning and model parameter adjustment, closing optional feedback loops toward the estimators. Remote interaction is provided through a BVOLT REST API, which exposes read access to system states and accepts write commands over HTTP from external clients, persisting all interactions through the same InfluxDB layer. Finally, the digital-twin outputs are visualized in a Grafana dashboard, providing a consolidated and remotely accessible view of SOC evolution and system behavior.

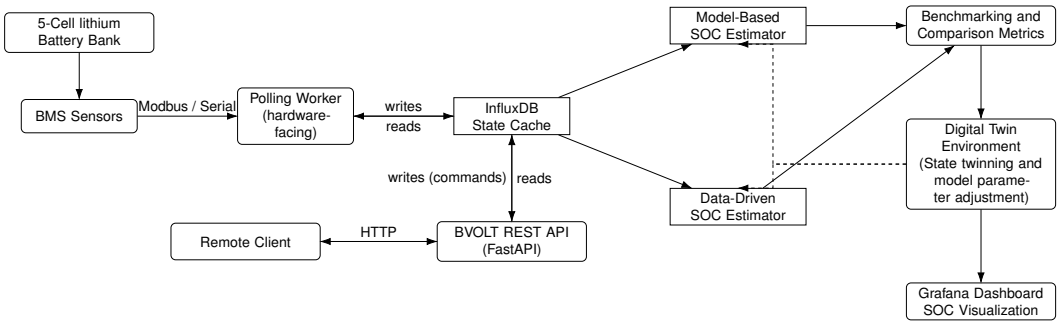

Figure 3 – Integrated architecture combining SOC estimation, Digital Twin modeling, and the BVOLT platform for remote monitoring and control.

6 CHRONOGRAM

This work is divided in 4 parts—work plan finalization; dissertation writing; software development; and important dates. The chronogram starts at 2026-01-02 with the technical core development. It is composed by BVOLT standard functions—telemetry data gathering and MODBUS communication interface, alongside with remotely accessible InfluxDB and Grafana interactions—, battery bank and BMS DTs building processes, and, at last, their validation. During the whole development phase, software documentation will be written and GitHub versioning will be conducted. The development is expected to end at the end of may 2026. The work plan document of the project shall be finalized at the beginning of february 2026; the activities that will take place after this are composed of the preparations for the qualification processed, to take place at 2026-02-25. After the qualification, the process of dissertation writing starts and shall end close to the beginning of june 2026, before the final defense. Article writing will happen amidst the of the review of the qualification presentation. The qualification, article writing and final defense compose all three important dates for this project. The chronogram can be seen at Figure 4.

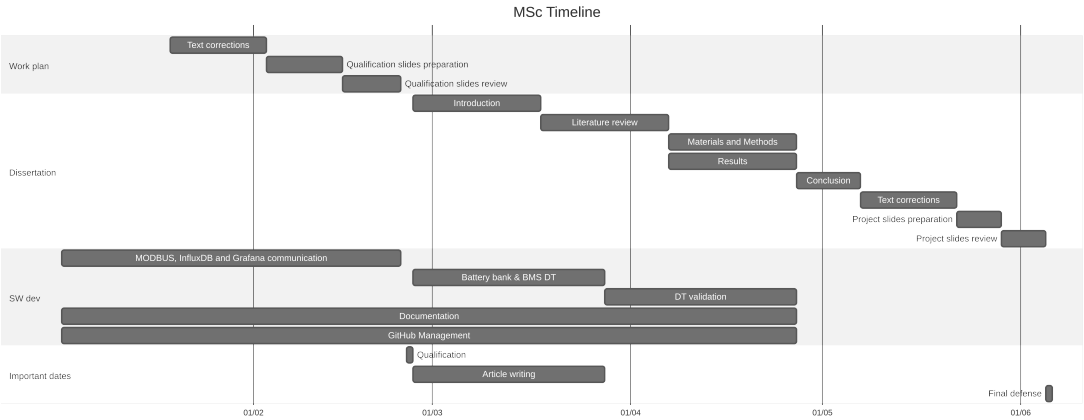

Figure 4 – The project work plan chronogram for the remaining of the year of 2026.

7 PERIODICALS AND SCIENTIFIC EVENTS

This section lists few periodicals and scientific events in which project-related papers and resumes can be published and presented.

7.1 PERIODICALS (JOURNALS)

1. Renewable and Sustainable Energy Reviews
   - Publisher: Elsevier
   - Impact Factor (2025): 16.3
   - Focus: A leading journal for review articles on sustainable energy, making it an excellent venue for a comprehensive paper derived from your literature review and project outcomes.
2. Applied Energy
   - Publisher: Elsevier
   - Impact Factor (2021): 11.446
   - Focus: Covers a broad range of topics in energy engineering, including energy conversion, conservation, and the optimization of energy processes, which aligns well with the goals of your MGDT.
3. IEEE Transactions on Industrial Informatics
   - Publisher: IEEE
   - Impact Factor (2024): 9.9
   - CiteScore: 22.5
   - Focus: This Q1 journal is highly relevant as it bridges industrial applications with information technology. Your project's focus on Industry 4.0 concepts, digital twins, IoT, and CPS fits perfectly within its scope.
4. IEEE Transactions on Smart Grid
   - Publisher: IEEE

- Impact Factor: 8.6
- Research Impact Score: 13.2
- Focus: A top-tier journal directly addressing the core topics of your project, including microgrids, renewable energy, energy management, and mathematical optimization.

5. IEEE Transactions on Sustainable Energy
   - Publisher: IEEE
   - Impact Factor: 8.6
   - Research Impact Score: 9.4
   - Focus: Excellent for papers emphasizing renewable energy aspects, control theory, and mathematical optimization within power systems.
6. IEEE Transactions on Power Systems
   - Publisher: IEEE
   - Journal Impact Factor (JIF): 7.2
   - Focus: A prestigious journal for foundational research in electric power systems. Your work on modeling, simulation, and control of a microgrid would be a strong candidate.
7. **Energies**
   - Publisher: MDPI
   - Focus: A peer-reviewed, open-access journal covering a wide range of topics in energy research and technology development. It offers a broad platform for interdisciplinary work like yours.

7.2 SCIENTIFIC EVENTS (CONFERENCES)

1. IEEE Power & Energy Society General Meeting (PES GM)
   - Description: Considered a premier power engineering conference, it

brings together academics and industry professionals from around the world. The scope is broad, covering all aspects of power and energy.

- Relevance: Ideal for presenting the overall project, from the modeling of LabInovação's microgrid to the results of your control strategies.

2. IEEE Conference on Decision and Control (CDC)

   - Description: The premier conference for the science of systems and control. It gathers an international community to discuss new research, future developments, and applications relevant to decision-making and control.
   - Relevance: A perfect venue to present the theoretical aspects of your Model Predictive Control (MPC) and other optimization algorithms.

3. IFAC World Congress

   - Description: A triennial congress organized on a global scale that covers all well-defined themes of Control Engineering.
   - Relevance: Given the strong control and automation component of your project, this is a top-tier event to disseminate your findings to the international control systems community.

4. ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (ACM BuildSys)

   - Description: A highly selective, single-track forum for research on systems issues related to energy efficiency in the built environment, including buildings and smart cities.
   - Relevance: Your project's application to a laboratory microgrid, which is essentially a small-scale built environment, makes it a good fit for this conference's focus on IoT and CPS for energy systems.

5. Microgrid Knowledge Conference

- Description: This event is a key gathering for a wide range of stakeholders in the microgrid space, including developers, regulators, and large energy consumers.
- Relevance: An excellent venue for the more applied aspects of your work, showcasing the practical implementation and benefits of the BVOLT toolkit.

REFERENCES

ALLEN, M. R. *et al.* Framing and Context. *In:* MASSON-DELMOTTE, V. *et al.* (eds.). **Global Warming of 1.5℃. An IPCC Special Report on the impacts of global warming of 1.5℃ above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change, sustainable development, and efforts to eradicate poverty**. Cambridge, UK and New York, NY, USA: Cambridge University Press, 2018. p. 49–92. DOI: 10.1017/9781009157940.003. Cit. on p. 21.

ANGELOVA, Dorotea Dimitrova *et al.* A Review on Digital Twins and Its Application in the Modeling of Photovoltaic Installations. **Energies**, v. 17, n. 5, 2024. ISSN 1996-1073. DOI: 10.3390/en17051227. Available from: <https://www.mdpi.com/1996-1073/17/5/1227>. Cit. on pp. 30, 34, 37.

ASTRAL. **uv Documentation**. [*S. l.: s. n.*], 2024. Available from: <https://docs.astral.sh/uv/>. Visited on: 22 Dec. 2025. Cit. on p. 25.

BAZMOHAMMADI, Najmeh *et al.* Microgrid Digital Twins: Concepts, Applications, and Future Trends. **IEEE Access**, v. 10, p. 2284–2302, 2022. DOI: 10.1109/ACCESS.2021.3138990. Cit. on pp. 28, 29, 31–34.

CONSELHO NACIONAL DE DESENVOLVIMENTO CIENTÍFICO E TECNOLÓGICO (CNPQ). **Institutos Nacionais de Ciência e Tecnologia (INCT)**. [*S. l.: s. n.*], 2025. http://inct.cnpq.br/home/. Accessed on 16 September 2025. Cit. on p. 22.

DANILCZYK, William; SUN, Yan; HE, Haibo. ANGEL: An Intelligent Digital Twin Framework for Microgrid Security. *In:* 2019 North American Power Symposium (NAPS). [*S. l.: s. n.*], 2019. p. 1–6. DOI: 10.1109/NAPS46351.2019.9000371. Cit. on pp. 33, 34, 36.

DAS, Opy *et al.* Advancements in digital twin technology and machine learning for energy systems: A comprehensive review of applications in smart grids, renewable energy, and electric vehicle optimisation. **Energy Conversion and Management: X**, v. 24, p. 100715, 2024. ISSN 2590-1745. DOI: https://doi.org/10.1016/j.ecmx.2024.100715. Available from:

<https://www.sciencedirect.com/science/article/pii/S2590174524001934>. Cit. on pp. 28, 30, 32, 33, 35–40.

EATY, Naga Durga Krishna Mohan; BAGADE, Priyanka. Digital twin for electric vehicle battery management with incremental learning. **Expert Systems with Applications**, v. 229, p. 120444, 2023. ISSN 0957-4174. DOI: https://doi.org/10.1016/j.eswa.2023.120444. Available from: <https://www.sciencedirect.com/science/article/pii/S0957417423009466>. Cit. on pp. 30, 31.

FENG, Zhiming *et al.* Review of Digital Twin Technology Applications in Hydrogen Energy. **CHAIN**, v. 1, n. 1, p. 54–74, 2024. DOI: 10.23919/CHAIN.2024.000001. Cit. on pp. 29–31, 37.

FOLGADO, Francisco Javier *et al.* Review of Industry 4.0 from the Perspective of Automation and Supervision Systems: Definitions, Architectures and Recent Trends. **Electronics**, v. 13, n. 4, 2024. ISSN 2079-9292. DOI: 10.3390/electronics13040782. Available from: <https://www.mdpi.com/2079-9292/13/4/782>. Cit. on p. 28.

GILLERMAN, J.; FALK, H.; MACKIEWICZ, R. Focus on IEC TC 57 - power system reliability and profitability. **IEEE Power and Energy Magazine**, v. 3, n. 4, p. 66–67, 2005. DOI: 10.1109/MPAE.2005.1458233. Cit. on p. 35.

GRAFANA LABS. **Grafana**. [*S. l.: s. n.*], 2025. https://grafana.com/. Accessed: September 17, 2025. Cit. on p. 25.

INFLUXDATA INC. **InfluxData**. [*S. l.: s. n.*], 2025. https://www.influxdata.com/. Accessed: September 17, 2025. Cit. on p. 25.

JAMAL, Saif; TAN, Nadia M. L.; PASUPULETI, Jagadeesh. A Review of Energy Management and Power Management Systems for Microgrid and Nanogrid Applications. **Sustainability**, v. 13, n. 18, 2021. ISSN 2071-1050. DOI: 10.3390/su131810331. Available from: <https://www.mdpi.com/2071-1050/13/18/10331>. Cit. on p. 21.

JAVAID, Mohd; HALEEM, Abid; SUMAN, Rajiv. Digital Twin applications toward Industry 4.0: A Review. **Cognitive Robotics**, v. 3, p. 71–92, 2023. ISSN

2667-2413. DOI: https://doi.org/10.1016/j.cogr.2023.04.003. Available from: <https://www.sciencedirect.com/science/article/pii/S2667241323000137>. Cit. on p. 39.

JONES, David *et al.* Characterising the Digital Twin: A systematic literature review. **CIRP Journal of Manufacturing Science and Technology**, v. 29, p. 36–52, 2020. ISSN 1755-5817. DOI: https://doi.org/10.1016/j.cirpj.2020.02.002. Available from: <https://www.sciencedirect.com/science/article/pii/S1755581720300110>. Cit. on pp. 32, 39.

JOSHAL, Karan Singh; GUPTA, Neeraj. Microgrids with Model Predictive Control: A Critical Review. **Energies**, v. 16, n. 13, 2023. ISSN 1996-1073. DOI: 10.3390/en16134851. Available from: <https://www.mdpi.com/1996-1073/16/13/4851>. Cit. on p. 21.

KUMARI, Namita *et al.* A Comprehensive Review of Digital Twin Technology for Grid-Connected Microgrid Systems: State of the Art, Potential and Challenges Faced. **Energies**, v. 16, n. 14, 2023. ISSN 1996-1073. DOI: 10.3390/en16145525. Available from: <https://www.mdpi.com/1996-1073/16/14/5525>. Cit. on pp. 21, 22, 28–31, 33, 34, 36, 39, 40.

LEI, Zhongcheng *et al.* Digital twin based monitoring and control for DC-DC converters. **Nature Communications**, Nature Publishing Group, v. 14, n. 1, p. 5604, 2023. DOI: 10.1038/s41467-023-41248-z. Available from: <https://www.nature.com/articles/s41467-023-41248-z>. Cit. on p. 31.

LIU, Xin *et al.* A systematic review of digital twin about physical entities, virtual models, twin data, and applications. **Advanced Engineering Informatics**, v. 55, p. 101876, 2023. ISSN 1474-0346. DOI: https://doi.org/10.1016/j.aei.2023.101876. Available from: <https://www.sciencedirect.com/science/article/pii/S1474034623000046>. Cit. on pp. 29, 31, 32, 34.

MACHADO, Diogo Ortiz *et al.* Digital twin of a Fresnel solar collector for solar cooling. **Applied Energy**, v. 339, p. 120944, 2023. ISSN 0306-2619. DOI: 10.1016/j.apenergy.2023.120944. Available from:

<https://www.sciencedirect.com/science/article/pii/S0306261923003082>. Cit. on p. 32.

MACHADO, Diogo Ortiz *et al.* Digital twin of an absorption chiller for solar cooling. **Renewable Energy**, v. 208, p. 36–51, 2023. DOI: 10.1016/j.renene.2023.03.048. Available from: <https://www.sciencedirect.com/science/article/pii/S0960148123003403>. Cit. on p. 32.

MARTIN, Robert C. **Clean Architecture: A Craftsman's Guide to Software Structure and Design**. Upper Saddle River, NJ: Prentice Hall, 2017. (Robert C. Martin Series). Cit. on p. 25.

MARTIN, Robert C. **Clean Code: A Handbook of Agile Software Craftsmanship**. Upper Saddle River, NJ: Prentice Hall, 2008. (Robert C. Martin Series). Cit. on p. 25.

MISHRA, J.; SHANKAR, A. Optimizing wind-PV-battery microgrids for sustainable and resilient residential communities. **Scientific Reports**, Nature Publishing Group, v. 15, p. 24339, 2025. DOI: 10.1038/s41598-025-06354-6. Available from: <https://doi.org/10.1038/s41598-025-06354-6>. Cit. on p. 21.

OLIVEIRA LUNA, José Diogo Forte de *et al.* A novel joint energy and demand management system for smart houses based on model predictive control, hybrid storage system and quality of experience concepts. **Applied Energy**, v. 369, p. 123466, 2024. DOI: 10.1016/j.apenergy.2024.123466. Available from: <https://doi.org/10.1016/j.apenergy.2024.123466>. Cit. on pp. 31, 33.

PYTHON SOFTWARE FOUNDATION. **Python Programming Language**. [*S. l.: s. n.*], 2025. https://www.python.org/. Accessed: September 17, 2025. Cit. on p. 24.

RASHEED, Adil; SAN, Omer; KVAMSDAL, Trond. Digital Twin: Values, Challenges and Enablers From a Modeling Perspective. **IEEE Access**, v. 8, p. 21980–22012, 2020. DOI: 10.1109/ACCESS.2020.2970143. Cit. on pp. 32, 33, 36–39.

SAHOO, Buddhadeva *et al.* Digital twin enabled smart microgrid system for complete automation: An overview. **Results in Engineering**, v. 25, p. 104010, 2025. ISSN 2590-1230. DOI: https://doi.org/10.1016/j.rineng.2025.104010. Available from: <https://www.sciencedirect.com/science/article/pii/S2590123025000982>. Cit. on pp. 30, 35, 36, 38–40.

SEGOVIA, Mariana; GARCIA-ALFARO, Joaquin. Design, Modeling and Implementation of Digital Twins. **Sensors**, v. 22, n. 14, 2022. ISSN 1424-8220. DOI: 10.3390/s22145396. Available from: <https://www.mdpi.com/1424-8220/22/14/5396>. Cit. on pp. 36, 38.

SHAHZAD, Sulman *et al.* Model Predictive Control Strategies in Microgrids: A Concise Revisit. **IEEE Access**, v. 10, p. 122211–122225, 2022. DOI: 10.1109/ACCESS.2022.3223298. Cit. on p. 21.

SINGH, Maulshree *et al.* Digital Twin: Origin to Future. **Applied System Innovation**, v. 4, n. 2, 2021. ISSN 2571-5577. DOI: 10.3390/asi4020036. Available from: <https://www.mdpi.com/2571-5577/4/2/36>. Cit. on pp. 34, 35, 40, 41.

SMITH, Edward; ROBINSON, Duane; AGALGAONKAR, Ashish. Cooperative Control of Microgrids: A Review of Theoretical Frameworks, Applications and Recent Developments. **Energies**, v. 14, n. 23, 2021. ISSN 1996-1073. DOI: 10.3390/en14238026. Available from: <https://www.mdpi.com/1996-1073/14/23/8026>. Cit. on pp. 35, 38.

TAO, Fei *et al.* Digital twin modeling. **Journal of Manufacturing Systems**, v. 64, p. 372–389, 2022. ISSN 0278-6125. DOI: https://doi.org/10.1016/j.jmsy.2022.06.015. Available from: <https://www.sciencedirect.com/science/article/pii/S0278612522001108>. Cit. on pp. 31, 32, 37.

WEST, Timothy D.; BLACKBURN, Mark. Is Digital Thread/Digital Twin Affordable? A Systemic Assessment of the Cost of DoD's Latest Manhattan Project. **Procedia Computer Science**, v. 114, p. 47–56, 2017. Complex Adaptive Systems Conference with Theme: Engineering Cyber Physical Systems, CAS October 30 – November 1, 2017, Chicago, Illinois, USA. ISSN 1877-0509. DOI: https://doi.org/10.1016/j.procs.2017.09.003. Available

from:
<https://www.sciencedirect.com/science/article/pii/S1877050917317970>.
Cit. on p. 34.